

AWS EBS (Elastic Block Store) - Service Overview

Introduction

Amazon Elastic Block Store (EBS) provides persistent, block-level storage volumes for use with EC2 instances. Each EBS volume is automatically replicated within its Availability Zone to protect against component failure, offering high availability and durability distinct from the ephemeral instance store volumes that are physically attached to an EC2 host.

Volume Types

- General Purpose SSD (gp3): the default recommendation for most workloads, offering a baseline of 3,000 IOPS and 125 MB/s throughput independent of volume size, with the ability to provision additional IOPS and throughput separately at extra cost.
- Provisioned IOPS SSD (io1/io2): designed for I/O-intensive workloads such as large relational or NoSQL databases, supporting up to 256,000 IOPS on io2 Block Express volumes.
- Throughput Optimized HDD (st1): low-cost magnetic storage designed for frequently accessed, throughput-intensive workloads such as big data and log processing.
- Cold HDD (sc1): the lowest-cost HDD volume, designed for less frequently accessed workloads.

Snapshots

EBS snapshots are point-in-time, incremental backups of a volume, stored durably in Amazon S3 (though not directly accessible or listed as S3 objects). Because snapshots are incremental, only the blocks that have changed since the last snapshot are saved, reducing both storage cost and the time required to create the snapshot. Snapshots can be used to create new volumes, migrate data between Availability Zones or regions, or restore a volume to a previous state.

Availability Zone Scoping

An EBS volume exists within a single Availability Zone and can only be attached to EC2 instances within that same Availability Zone. To attach a volume's data to an instance in a different AZ, you must first create a snapshot and then create a new volume from that snapshot in the target AZ.

Encryption

EBS supports encryption at rest using AWS KMS. Encrypting a volume also encrypts any snapshots created from it, and any volumes subsequently created from those encrypted snapshots, without requiring any application-level changes, since encryption and decryption happen transparently at the storage layer.

Multi-Attach

Certain io1/io2 volume types support Multi-Attach, allowing a single EBS volume to be concurrently attached to multiple EC2 instances within the same Availability Zone. This requires cluster-aware file systems or applications capable of managing concurrent write access, since standard file systems are not designed for this.

Elastic Volumes

EBS Elastic Volumes allow you to dynamically increase volume size, adjust performance (IOPS/throughput), or change the volume type without detaching the volume or restarting the associated instance, minimizing downtime for capacity changes.

EBS vs Instance Store

Instance store volumes are physically attached to the host machine running an EC2 instance, offering very high, low-latency IOPS, but their data is ephemeral: it is lost if the instance is stopped, terminated, or the underlying hardware fails. EBS volumes, by contrast, are network-attached, independent storage resources that persist regardless of the instance's state, and can be detached from one instance and reattached to another. The tradeoff is that EBS introduces a small amount of network latency compared to physically local instance store storage.

Pricing Example

A 100 GiB gp3 volume costs approximately \$0.08 per GiB-month, or about \$8/month, and includes the baseline 3,000 IOPS and 125 MB/s throughput at no extra charge. Provisioning additional IOPS beyond the baseline (up to 16,000 IOPS on gp3) costs an incremental per-IOPS rate. By comparison, an io2 volume provisioned for very high, consistent IOPS (for example, 10,000 IOPS) costs substantially more per GiB, reflecting its design for latency-sensitive, mission-critical database workloads rather than general-purpose storage.

Common Use Cases

- Root volumes for EC2 instances, allowing the instance to be stopped and started without losing data
- Database storage for self-managed databases running on EC2
- Storage backing RDS instances (RDS itself uses EBS internally)
- Big data analytics workloads requiring high sequential throughput

Best Practices

- Take regular automated snapshots of critical volumes using AWS Backup or Data Lifecycle Manager.
- Choose gp3 over the older gp2 volume type for better baseline performance at lower cost in most cases.
- Enable EBS encryption by default at the account level for compliance and security posture.
- Monitor VolumeQueueLength and burst balance metrics in CloudWatch to detect IOPS bottlenecks before they impact application performance.